Negating conditionals:-

$$\sim(P \Rightarrow Q)$$

$$P \wedge \sim Q$$

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

1st way:- $\sim(P \Rightarrow Q) = P \wedge \sim Q$

2nd way: $\sim(P \Rightarrow Q) = \sim(\sim P \vee Q) = \sim(\sim P) \wedge \sim Q$

$$P \wedge \sim Q$$

ex:- R:- if $\underbrace{a \text{ is odd}}_{P}$, then $\underbrace{a^2 \text{ is odd}}_{Q}$.

$\sim R = ?$

$$P \Rightarrow Q$$

$$\sim(P \Rightarrow Q) = P \wedge \sim Q$$

$\sim R$:- a is odd and a^2 is not odd